

(Giommi et al. 2012; Wan et al. 2024), mass accretion rate (Boula et al. 2019, 2026) or beaming effects (Fan et al. 2017; Yang et al. 2022c; Ouyang et al. 2023). The location of the ν_{peak} also classifies RLAGNs into low-synchrotron-peaked (LSP, $\log(\nu_{\text{peak}}/\text{Hz}) < 14$), intermediate-synchrotron-peaked (ISP, $14 \leq \log(\nu_{\text{peak}}/\text{Hz}) < 15$), high-synchrotron-peaked (HSP, $15 \leq \log(\nu_{\text{peak}}/\text{Hz}) \leq 17$), and extreme high-synchrotron-peaked (EHSP, $\log(\nu_{\text{peak}}/\text{Hz}) \geq 17$) sources (Abdo et al. 2010c; Fan et al. 2016; Láinez et al. 2025; Sibani et al. 2026; Abe et al. 2026). In general, FSRQs are homogeneous and mostly correspond to LSPs, whereas BL Lacs are heterogeneous, including EHBLs, HBLs, IBLs, and LBLs (extreme high-, high-, intermediate-, and low-synchrotron-peaked BL Lacs) (Costamante et al. 2001; Nieppola et al. 2006; Foffano et al. 2019; Ajello et al. 2022; Yang et al. 2022b; Lian et al. 2026).

Since the launch of the *Fermi* Large Area Telescope (hereafter *Fermi*), blazars have been detected to dominate the extragalactic γ -ray sky (Abdo et al. 2010b; Abdollahi et al. 2022). The latest Third Data Release of the Fourth *Fermi* AGN Catalogue (4LAC-DR3) contains 3814 γ -ray detected AGNs, among which 3743 AGNs (98.14%) are classified as blazars, including 792 FSRQs, 1458 BL Lacs, and 1493 blazar candidates of uncertain type (BCUs) (Ajello et al. 2022; Abdollahi et al. 2022). Their γ -ray emissions are closely linked to their radio properties, exhibiting higher radio brightness temperature, stronger core-dominance phenomena, significant apparent superluminal motions, strongly polarized jets and larger Doppler factors (Moellenbrock et al. 1996; Lister & Homan 2005; Kovalev et al. 2009; Lister et al. 2009; Linford et al. 2012; Wu et al. 2014; Pei et al. 2020; Homan et al. 2021; Xiao et al. 2022). The strong correlation between radio and γ -ray emissions in blazars has been studied, showing that the high-energy output is closely linked to the relativistic jets responsible for compact radio emissions (Fan et al. 1998; Fossati et al. 1998; Kovalev et al. 2009; Ojha et al. 2010; Lister et al. 2011; Fan et al. 2012; Linford et al. 2012; Böck et al. 2016; Cui et al. 2026). Very Long Baseline Interferometry (VLBI) provides the highest angular resolution, which can help further resolve the inner jets of blazars and measure their jet kinematics (Deller & Middelberg 2014; Böck et al. 2016; Petrov & Kovalev 2025). The connection between VLBI and *Fermi* motivates one to study the jet properties of γ -ray blazars, and understand their high-energy radiation mechanisms. In this paper, we utilised a large sample combining the RFC (Radio Fundamental Catalogue) VLBI catalogue¹ and the *Fermi* 4LAC-DR3 AGN catalogue (Ajello et al. 2022) to discuss the radio and γ -ray properties, as well as the high-energy mechanisms.

2 SAMPLE

To investigate the parsec-scale radio properties of γ -ray blazars, we cross-matched the 4LAC-DR3 sample with the RFC catalogue compiled by Petrov & Kovalev (2025), who present flux density measurements for 21942 compact radio sources based on the reanalysis of 1088 VLBI experiments conducted between 1994 and 2024. The observations cover the precise positions and the flux densities at 2.2–2.4 GHz (S band), 4.1–5 GHz (C band), 7.3–8.8 GHz (X band), 15.2–15.5 GHz (U band), and 22–24.2 GHz (K band). There are three different baseline lengths for the VLBI detection: median flux density at the baseline projection lengths shorter than 1000 km, between 1000 km and 5000 km, and longer than 5000 km.

The cross-matching process between the *Fermi* 4FGL-DR3 and

RFC catalogues yields 2578 γ -ray blazars, including 1057 BL Lacs, 735 FSRQs, and 786 BCUs. The VLBI flux densities are obtained at C band or X band with baseline projections longer than 5000 km, which corresponds to the unresolved flux and refers to the compact core emission on the parsec scale for blazars. This results in a blazar sample of confirmed 976 BL Lacs and 711 FSRQs with the available VLBI radio core emissions. The compact VLBI radio spectrum for RLAGNs in the GHz band is flat due to synchrotron self-absorption (e.g., Blandford & Königl 1979; Lobanov 1998; Abdo et al. 2010c; Grandi et al. 2025); therefore, these measurements between C band (4.1 – 5 GHz) and X band (7.3 – 8.8 GHz) provide a reliable and consistent compact radio core emission for blazars.

Overall, the sample includes 976 BL Lacs and 711 FSRQs with both the VLBI and γ -ray parameters. Although the sample is not statistically complete from both the VLBI and *Fermi* surveys, compared with earlier radio and γ -ray studies based on a few hundred sources (e.g., Lister et al. 2011; Linford et al. 2012; Böck et al. 2016), its large sample size and broad coverage in radio flux, γ -ray flux, ν_{peak} , and spectral properties make it representative of the bright γ -ray blazar population. Therefore, selection biases are not expected to strongly affect our results.

2.1 Radio and γ -ray properties

The VLBI observations cover a wide range in flux density for the 1687 blazars, from $S_{\text{VLBI}} = 0.004$ Jy to $S_{\text{VLBI}} = 9$ Jy. The γ -ray average energy flux is taken from the 4FGL-DR3, which is integrated over the 0.1–100 GeV band (Ajello et al. 2022), ranging from $\log f_{\gamma} = -9.07$ erg/cm²/s to $\log f_{\gamma} = -12.43$ erg/cm²/s for 1687 blazars.

The average radio and γ -ray emissions for FSRQs ($S_{\text{VLBI}} = 0.41 \pm 0.03$ Jy; $\log f_{\gamma} = -11.18 \pm 0.02$ erg/cm²/s) are higher than those of BL Lacs ($S_{\text{VLBI}} = 0.10 \pm 0.01$ Jy; $\log f_{\gamma} = -11.30 \pm 0.01$ erg/cm²/s), with the clear differences determined by the Kolmogorov–Smirnov (K-S) tests of the probability of $p < 0.05$. Figure 1 shows the relation between the VLBI radio flux and the γ -ray energy flux. This clear positive correlation is confirmed by the Kendall τ test correlation coefficients of $r = 0.48$ and $p < 10^{-4}$ for BL Lacs, and $r = 0.47$ and $p < 10^{-4}$ for FSRQs.

Lister et al. (2009) discussed that the fainter VLBA² radio luminosities of the BL Lacs require a higher intrinsic (unbeamed) γ -ray to radio luminosity ratio than FSRQs, in order to account for their higher *Fermi*-detection rate (48% detection rate for BL Lacs, while 20% for FSRQs in the MOJAVE³ sample). With the large sample, the 976 BL Lacs show a lower average VLBI radio emission than the 711 FSRQs, in line with the idea of Lister et al. (2009) that a higher intrinsic (unbeamed) γ -ray to radio luminosity ratio might be involved for BL Lacs' detection, which could be further investigated by the unified scenario between blazars and radio galaxies in the future.

2.2 γ -ray loudness and photon index

The compact radio core emission traces the synchrotron emission from the relativistic jet, while the γ -ray emission is generally attributed to IC processes (Ghisellini et al. 2011; Cerruti 2020). Differences in dominant radiative mechanisms (e.g., SSC versus EC) are

¹ https://astrogeo.org/sol/rfc/rfc_2024d (Petrov & Kovalev 2025).

² Very Long Baseline Array.

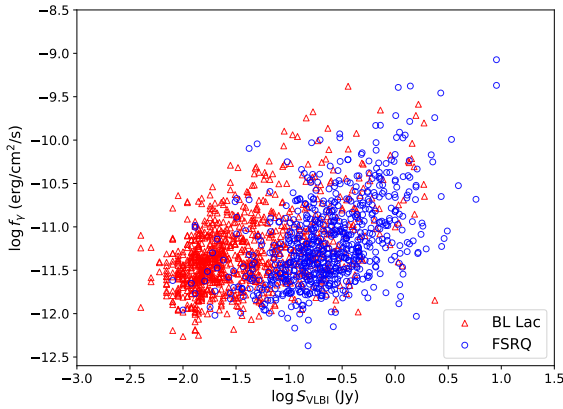
³ Monitoring Of Jets in Active galactic nuclei with VLBA Experiments.

Table 1. Radio and γ -ray properties for 1687 *Fermi* blazars.

Source Name (1)	Class (2)	$\log \nu_{\text{peak}}$ (3)	Γ_γ (4)	$\log f_\gamma$ (5)	VLBI Counterpart (6)	$\log S_{\text{VLBI}}$ (7)	$\log G_r$ (8)
4FGL J0001.2-0747	BL Lac	13.96	2.12	-11.08	RFC J0001-0746	-1.29	3.51
4FGL J0001.5+2113	FSRQ	14.20	2.65	-10.59	RFC J0001+2113	-1.18	3.89
4FGL J0003.2+2207	BL Lac	13.92	2.12	-12.19	RFC J0003+2204	-2.10	3.20
4FGL J0003.9-1149	BL Lac	12.39	2.18	-11.63	RFC J0004-1148	-0.69	2.37
4FGL J0004.0+0840	BL Lac	14.48	2.00	-11.94	RFC J0003+0841	-1.89	3.25
4FGL J0004.3+4614	FSRQ	12.41	2.59	-11.29	RFC J0004+4615	-0.81	2.83
4FGL J0004.4-4737	FSRQ	13.12	2.44	-11.27	RFC J0004-4736	-0.68	2.71
4FGL J0005.9+3824	FSRQ	13.12	2.62	-11.19	RFC J0005+3820	-0.92	3.03
4FGL J0006.3-0620	BL Lac	12.92	2.15	-11.85	RFC J0006-0623	0.37	1.08
4FGL J0008.0+4711	BL Lac	13.52	2.05	-10.73	RFC J0007+4712	-1.70	4.27
4FGL J0008.4-2339	BL Lac		1.72	-11.67	RFC J0008-233A	-1.82	3.46
4FGL J0009.1+0628	BL Lac	13.39	2.15	-11.23	RFC J0009+0628	-1.03	3.10
4FGL J0009.3+5030	BL Lac	15.10	1.99	-10.64	RFC J0009+5030	-2.05	4.71
4FGL J0010.6+2043	FSRQ	12.56	2.39	-11.71	RFC J0010+2047	-1.28	2.87
4FGL J0010.6-3025	FSRQ	12.99	2.43	-11.27	RFC J0010-3027	-0.73	2.76
4FGL J0011.4+0057	FSRQ	12.84	2.35	-11.13	RFC J0011+0057	-0.91	3.08
4FGL J0013.0+3355	FSRQ	12.50	2.26	-11.75	RFC J0012+3353	-0.89	2.44
4FGL J0013.1-3955	BL Lac	12.86	2.05	-11.38	RFC J0012-3954	-0.31	2.22
...	...	...	...	...	...	...	...

Note: Columns list (1) source name, (2) class, (3) logarithmic synchrotron peak frequency (ν_{peak}), (4) γ -ray photon index (Γ_γ), (5) logarithmic gamma-ray energy flux, $\log f_\gamma$ (in units of $\text{erg}/\text{cm}^2/\text{s}$), (6) VLBI counterpart, (7) logarithmic VLBI flux density, $\log S_{\text{VLBI}}$ (in units of Jy), and (8) logarithmic ratio $G_r = f_\gamma/\nu S_{\text{VLBI}}$.

A portion of this table is shown here for guidance. The full table is available in Zenodo, at <https://doi.org/10.5281/zenodo.21695209> [doi].

**Figure 1.** γ -ray energy flux (f_γ) versus VLBI radio flux density (S_{VLBI}) for *Fermi* blazars. Red triangles represent BL Lacs and blue circles represent FSRQs.

expected to produce distinct γ -ray loudness distributions. The γ -ray loudness is defined as in [Lister et al. \(2011\)](#); [Linford et al. \(2012\)](#),

$$G_r = \frac{\nu L_\gamma}{\nu L_{\text{VLBI}}} = \frac{f_\gamma}{\nu S_{\text{VLBI}}}, \quad (1)$$

where L_γ and L_{VLBI} are the γ -ray and VLBI luminosities, respectively. Here f_γ is the γ -ray energy flux and S_{VLBI} is the monochromatic radio flux density. Because the radio and γ -ray luminosities scale similarly with luminosity distance, G_r is insensitive to redshift effects. This makes it particularly useful for BL Lacs with uncertain or missing redshift measurements, which are often excluded from samples requiring redshift information (e.g., [Fan & Wu 2018](#); [Láinez et al. 2025](#)). Cross-matching the redshift from the 4LAC-DR3 catalogue shows that 357 out of 976 BL Lacs would be excluded due

to the lack of redshift measurements. As a result, our sample includes a larger fraction of BL Lacs without measured redshifts than previous studies. This allows for a more complete comparison between BL Lacs and FSRQs in both radio and γ -ray properties.

All sources in our sample are γ -ray loud, with $\log G_r$ spanning $\log G_r = 1.07$ to $\log G_r = 4.70$ for 1687 blazars. [Linford et al. \(2012\)](#) showed a higher range of $\log G_r = 2.74$ – 5.27 for the 1FGL (*Fermi* First Source Catalogue) sample. This difference likely arises from the different temporal coverage of the γ -ray data. The 1FGL catalogue is biased toward bright sources, whereas the 12-year averaged data would smooth out strong variability and reduce the contribution from flaring states, leading to lower mean γ -ray fluxes and smaller G_r values.

[Lister et al. \(2011\)](#) reported a clear separation in the distribution of G_r between BL Lacs and FSRQs, whereas [Linford et al. \(2012\)](#) found no difference between the two populations. [Cui et al. \(2026\)](#) adopted the latest Nanshan 26-m radio telescope to observe *Fermi* blazars and obtained a slightly higher γ -ray loudness in BL Lacs than FSRQs. In this sample, the FSRQs are less γ -ray loud, and BL Lacs show a slightly higher average γ -ray loudness. The average γ -ray loudness for BL Lacs ($\log G_r = 3.43 \pm 0.02$) is larger than that of FSRQs ($\log G_r = 2.75 \pm 0.02$), with a significant difference of the K-S probability $p < 0.05$.

Figure 2 presents the distribution between G_r and the γ -ray photon index Γ_γ . A significant anti-correlation is observed for both BL Lacs [$\log G_r = -(1.40 \pm 0.07)\Gamma_\gamma + (6.31 \pm 0.14)$ with $r = -0.55$ and $p < 10^{-4}$] and FSRQs [$\log G_r = -(0.67 \pm 0.09)\Gamma_\gamma + (4.40 \pm 0.21)$ with $r = -0.28$ and $p < 10^{-4}$], while BL Lacs preferentially occupy regions of higher γ -ray loudness and harder spectra, and FSRQs cluster at softer photon indices and lower G_r . [Böck et al. \(2016\)](#) investigated the relationship between G_r and Γ_γ using contemporaneous γ -ray and radio data, and reported a clear anti-correlation between the two quantities. Their result is consistent with earlier findings by [Linford et al. \(2012\)](#) and [Lister et al. \(2011\)](#). The phenomenon is in-

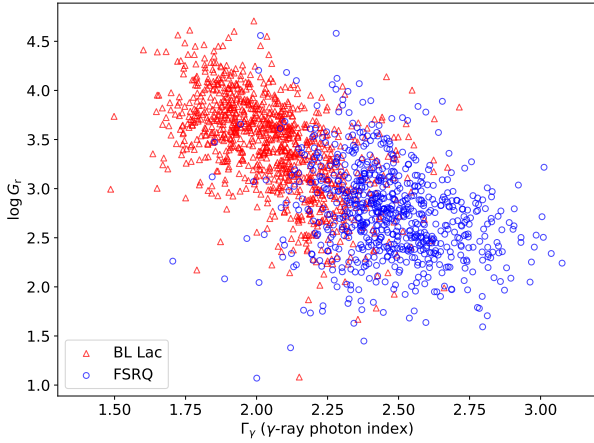


Figure 2. γ -ray loudness G_r versus γ -ray photon index Γ_γ for *Fermi* blazars. Symbols are the same as in Figure 1. BL Lacs occupy regions of higher G_r and harder spectra compared to FSRQs.

terpreted within the shift of the SED. As the synchrotron peak shifts to higher frequencies, the radio emission decreases, while the high-energy peak moves to higher energies, resulting in enhanced γ -ray emission and a harder spectrum (Böck et al. 2016; Fan & Wu 2018). This behaviour is supported by the anti-correlation between the radio emissions and ν_{peak} , namely the blazar sequence (Fossati et al. 1998) (also shown in Fig. 3): high- ν_{peak} sources are characterised by lower radio emissions. Higher radio emission (also higher radio core brightness temperature, e.g., Linford et al. 2012) is generally associated with larger Doppler factors (e.g., Fan et al. 2009; Hovatta et al. 2009; Lioudakis et al. 2018; Homan et al. 2021). This implies that LBLs are more strongly beamed in the radio band than HBLs.

Abdo et al. (2010b) presented that the Γ_γ is also anti-correlated with the ν_{peak} for the 1FGL blazars. Therefore, the relationship between γ -ray loudness and ν_{peak} is expected to be positively correlated, which has been discussed in the literature (Ojha et al. 2010; Lister et al. 2011; Linford et al. 2012). However, due to the sample limit, different studies report varying trends. The relation between the γ -ray loudness and ν_{peak} for a large sample is presented in Sec. 2.3.

2.3 γ -ray Loudness and synchrotron peak frequency

Ajello et al. (2022) derived the ν_{peak} for 1913 blazars (731 FSRQs and 1182 BL Lacs; see their Table 2) by fitting the synchrotron component of the SED with either a third-degree polynomial in the log-log plane or a parametric method based on broadband spectral indices. Among these sources, 1506 blazars (844 BL Lacs and 662 FSRQs) are matched in this paper. The distribution between ν_{peak} and G_r for 1506 blazars is shown in Figure 4. The dashed vertical lines in Figure 4 mark the boundaries between LSPs ($\log(\nu_{\text{peak}}/\text{Hz}) < 14$), ISPs ($14 \leq \log(\nu_{\text{peak}}/\text{Hz}) < 15$), and HSPs ($\log(\nu_{\text{peak}}/\text{Hz}) \geq 15$) (e.g., Abdo et al. 2010c; Fan et al. 2016; Yang et al. 2022b). As expected from the relation between the γ -ray photon index and the ν_{peak} (Abdo et al. 2010b; Ackermann et al. 2011b; Abdollahi et al. 2020), the total tendency is positive from the lower ν_{peak} and γ -ray loudness to the higher ν_{peak} and γ -ray loudness. FSRQs and LBLs exhibit a weak positive trend with substantial scatter. In contrast, the trend appears to flatten in HBLs (Fig. 4). This phenomenon may arise from the different high-energy processes operating in these subclasses, as well as the influence of the Klein–Nishina (KN) effect.

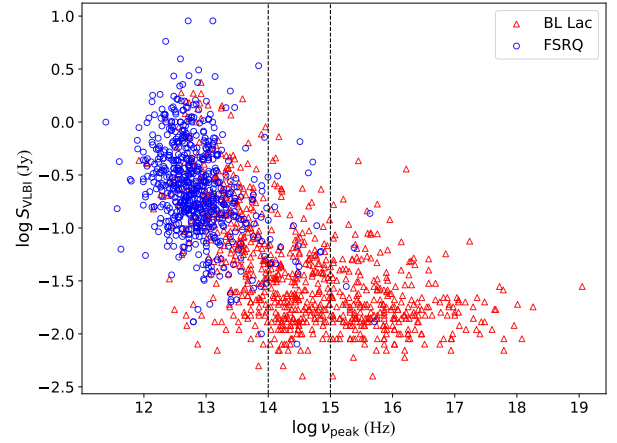


Figure 3. VLBI flux density (S_{VLBI}) versus synchrotron peak frequency ν_{peak} for *Fermi* blazars. The dashed vertical lines indicate the boundaries between LSP, ISP, and HSP blazars (LSP: $\log(\nu_{\text{peak}}/\text{Hz}) < 14$; ISP: $14 \leq \log(\nu_{\text{peak}}/\text{Hz}) < 15$; HSP: $\log(\nu_{\text{peak}}/\text{Hz}) \geq 15$). Symbols are the same as in Figure 1.

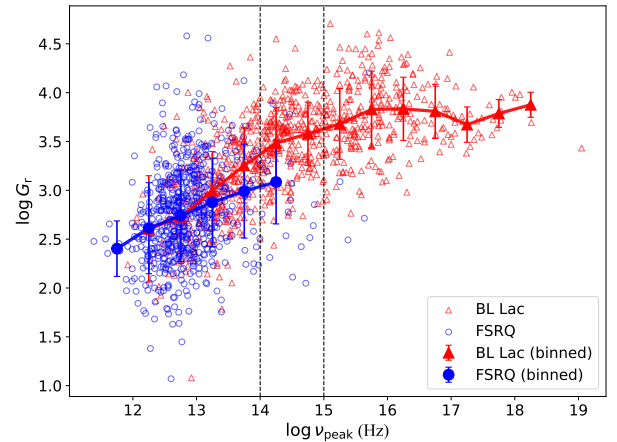


Figure 4. γ -ray loudness G_r versus synchrotron peak frequency ν_{peak} for *Fermi* blazars. The interval of $\log \nu_{\text{peak}}$ with a bin size of 0.5 for both BL Lacs and FSRQs. The dashed vertical lines indicate the boundaries between LSP, ISP, and HSP blazars (LSP: $\log(\nu_{\text{peak}}/\text{Hz}) < 14$; ISP: $14 \leq \log(\nu_{\text{peak}}/\text{Hz}) < 15$; HSP: $\log(\nu_{\text{peak}}/\text{Hz}) \geq 15$). Symbols are the same as in Figure 1.

3 DISCUSSION

The connection between radio and γ -ray emissions in blazars has been extensively studied, particularly, since the beginning of the *Fermi* era (Abdo et al. 2010b; Ackermann et al. 2011a; Linford et al. 2011; Chen & Bai 2011; Lister et al. 2011; Fan et al. 2012; Arshakian et al. 2012; Böck et al. 2016; Cui et al. 2026). The close relations between the radio and γ -ray luminosities (or fluxes) have been confirmed to support the link between the low-energy and high-energy bands. Chi & Young (1997) proposed that high-resolution VLBI data are more suitable when studying the beaming patterns through the γ -ray to radio flux ratio. Fan et al. (1998) found a close correlation between the γ -ray and the high-frequency (1.3 mm, 230 GHz) radio emissions for 44 γ -ray RLAGNs, which means that the γ -ray is associated with the radio emission from the jet. Huang et al. (1999) investigated the correlation between the flux ratio and the

Doppler factors (δ) for a sample of AGNs detected by EGRET⁴. Based on the SSC constraints proposed by Ghisellini et al. (1993), they estimated δ_{SSC} and found strong correlations between δ_{SSC} and both the γ -ray to optical flux ratio and the γ -ray to infrared flux ratio. These results favour an EC model, in which external seed photons contribute to the γ -ray emission in EGRET AGNs. However, Fan et al. (2017) found a tight correlation between γ -ray luminosity and radio luminosities in blazars, and proposed that an SSC emission is the origin for most γ -ray blazars because EC emission is expected to dilute any tight correlation between low- and high-energy emissions, while SSC emission should show tight correlations between them. In this paper, we compiled a large sample of *Fermi* blazars, and confirmed a statistically positive correlation between the VLBI radio flux and the γ -ray energy flux (Figure 1), as also shown in Abdo et al. (2010b); Pushkarev et al. (2010); Lister et al. (2011); Arshakian et al. (2012); Linford et al. (2012); Fan et al. (2012); Böck et al. (2016); Fan et al. (2017); Yang et al. (2022b). Such a correlation is generally interpreted as evidence that the radio and γ -ray emission originate from a one-zone emitting region within the relativistic jet.

As pointed out by Costamante & Ghisellini (2002); Tavecchio et al. (2010), the brightest γ -ray blazars are those with the largest radio flux. Therefore, the observed correlation between the radio and γ -ray bands is likely driven by Doppler beaming, which enhances both the radio and high-energy emissions, and their correlations (e.g., Fan et al. 1998; Lister et al. 2009); however, there may be an additional dependence on the high-energy radiation mechanism. A purely EC model would not predict a tight correlation between radio and γ -ray flux if the seed photons originate entirely outside the jet. Meanwhile, the δ_{EC} is not necessarily related to the δ_{syn} . In contrast, the SSC scenario links the synchrotron and IC emission through the same electron population and the similar inner jet Doppler factors $\delta_{\text{SSC}} \sim \delta_{\text{syn}}$, producing an intrinsic connection between radio and γ -ray output. Therefore, the clear correlation between radio and γ -ray emissions favours the one-zone leptonic model with the SSC radiation mechanism and suggests that the SSC mechanism contributes to both BL Lacs and FSRQs. The SSC contributions evidently support the SEDs for some FSRQs (e.g., Ghisellini et al. 1998, 2010; Böttcher et al. 2013; Yuan et al. 2022).

Source variability and the use of non-simultaneous data can weaken the observed correlation. The averaged γ -ray flux may smooth out short-term variability, and time lags between radio and γ -ray bands may introduce additional scatter. Nevertheless, the large sample size helps reveal the underlying statistical trend. Even in FSRQs, where EC emission dominates the high-energy output, the observed correlation suggests that SSC still contributes to the radio and γ -ray components.

3.1 SED shift for G_r and Γ_γ

The γ -ray emissions for FSRQs are explained by the EC, with the higher external photon density rapidly cooling down the relativistic high-energy electrons and steepening the spectrum (Ghisellini et al. 2017; Prandini & Ghisellini 2022; Ye et al. 2026). However, one would expect a softer γ -ray photon index and higher γ -ray emissions as presented in Ghisellini et al. (2011) and Ghisellini et al. (2017), where BL Lacs and FSRQs occupy different regions in the γ -ray luminosity-spectral index plane due to differences in jet power, accretion mode, and external photon fields. In this case, the stronger

γ -ray emission should enhance G_r in FSRQs, and a positive correlation between G_r and Γ_γ is expected from BL Lacs to FSRQs.

Lister et al. (2011) first presented the G_r and Γ_γ with an anti-correlation for both BL Lacs and FSRQs, which is confirmed and discussed by many authors (e.g., Linford et al. 2012; Fan et al. 2012; Böck et al. 2016; Fan & Wu 2018; Cui et al. 2026). With a large sample of 1687 blazars, we also found and confirmed a significant anti-correlation between G_r and Γ_γ (Figure 2). As ν_{peak} increases, the radio flux density tends to decrease (Figure 3), namely the blazar sequence (Fossati et al. 1998; Prandini & Ghisellini 2022), where weaker radio emission is observed in HSPs. Meanwhile, the shift of the high-energy SED component toward the *Fermi* band enhances the observed γ -ray emission, resulting in larger G_r values. However, for sources with $\log \nu_{\text{peak}} > 15$ (e.g., HSPs), the radio flux density exhibits little variation with increasing ν_{peak} . This behaviour may arise from the radio sensitivity limit and/or selection effects in the radio band. Overall, it is interesting to note that the contributions from the shift of the SED might be stronger than the effect of Compton cooling for both BL Lacs and FSRQs.

3.2 High-energy radiation mechanism for G_r and ν_{peak}

As mentioned before, the spectral index is anti-correlated with ν_{peak} in Abdo et al. (2010b) and Ackermann et al. (2011b). Therefore, the relationship between G_r and ν_{peak} is expected to be closely positively correlated. But limited by the sample size, different studies report inconsistent trends:

Lister et al. (2011) proposed a parameter of the γ -ray loudness, which is the ratio of average γ -ray luminosity during the first 11 months of the *Fermi* mission to the median 15 GHz VLBA radio luminosity, and showed the positive relation between the G_r and the ν_{peak} for 1FGL BL Lacs. One year later, Linford et al. (2012) adopted a larger sample of the *Fermi* blazars with both 1FGL blazars and 2FGL (*Fermi* Second Source Catalogue) blazars, but they obtained a different result: the mostly weak or tentative correlations for BL Lacs and FSRQs were found for the 1FGL blazars, but low-significance correlations for the 2FGL sample. When a second data set from the 2LAC (the *Fermi* Second Catalogue of AGN, Ackermann et al. 2011b), estimates of the ν_{peak} based on the empirical method (e.g., Abdo et al. 2010b), show even weaker correlation between the G_r and the ν_{peak} (Linford et al. 2012).

As shown in Fig. 3 of Linford et al. (2012), a handful of BL Lacs with the high ν_{peak} and relatively low G_r significantly affected the positive correlations between the G_r and ν_{peak} . They explained that the different results may arise from sample selection effects. The sample used by Linford et al. (2012) contains 30 HBLs, compared to only 17 HBLs in Lister et al. (2011). Limited by the small sample size, Lister et al. (2011) obtained a significant positive relation, which could be due to the lack of HBL samples.

The small sample at the beginning of the *Fermi* operation will preferably catch the bright sources in both the radio and γ -ray bands (e.g., Kovalev et al. 2009; Ojha et al. 2010). When a larger sample is considered in this paper, we found different trends for both FSRQs and BL Lacs, as well as the subclasses, LBLs, IBLs, and HBLs, as shown in Fig. 4. The relation for HBLs is flatter than both LBLs and IBLs, while LBLs and IBLs do present a tentatively positive relation between the γ -ray loudness and ν_{peak} . Therefore, when more HBLs are involved in the sample, the flat trend in HBLs might naturally weaken the positive relationship.

The different slopes observed in the G_r – ν_{peak} relation (Figure 4) among LBLs, IBLs, and HBLs provide important insight into the underlying radiation processes. As suggested by Linford et al. (2012),

⁴ The Energetic Gamma-Ray Experiment Telescope.

an SSC origin of the γ -ray emission would favour a strong correlation between G_r and ν_{peak} , since in this scenario the seed photons for IC scattering are produced internally via synchrotron emission, and both the synchrotron and γ -ray components share similar Doppler boosting factors. For FSRQs, however, the situation is more complex. The seed photon field is supplemented by external radiation components, such as those from the broad-line region or the dusty torus, leading to EC scattering. These external photon fields introduce additional Doppler boosting effects and variability in the γ -ray emission that are not directly linked to the synchrotron properties, thereby increasing the scatter in the G_r – ν_{peak} relation and diluting any intrinsic correlation.

In our sample of 662 FSRQs, the distribution is highly scattered, with most sources located at $\log(\nu_{\text{peak}}/\text{Hz}) < 14$, consistent with an EC-dominant contribution. Nevertheless, the binned data (with bin size $\Delta \log \nu_{\text{peak}} = 0.5$; see the binned blue circles in Fig. 4) reveal a weak but noticeable positive trend, in which G_r increases on average with ν_{peak} . This suggests that, although EC likely dominates the γ -ray emission in FSRQs, SSC may still contribute to the high-energy output and plays a role in shaping the positive correlation.

For BL Lacs, different behaviours emerge among the LBL, IBL, and HBL subclasses. Using the same binning scheme, LBLs and IBLs exhibit a clearer positive correlation between G_r and ν_{peak} . Meanwhile, the G_r of LBLs spans a wide range, from $\log G_r = 1.08$ to $\log G_r = 4.36$, making them as scattered as FSRQs in the G_r – ν_{peak} plane. This behaviour suggests that LBLs may resemble FSRQs, in which EC emission contributes significantly to the observed scatter, although SSC emission still plays an important role in producing the overall positive trend between G_r and ν_{peak} . In contrast, IBLs exhibit a relatively tighter distribution, possibly indicating a stronger contribution from SSC emission. The enlarged samples of LBLs and IBLs used in this work improve the statistical robustness compared to earlier studies (e.g., [Lister et al. 2011](#); [Linford et al. 2012](#)) and further support the importance of SSC emission in shaping the γ -ray loudness sequence.

The continuity of the G_r – ν_{peak} correlation for BL Lacs in [Lister et al. \(2011\)](#) suggests that the LBLs and HBLs belong to the same parent population. However, HBLs exhibit a much flatter distribution than LBLs in the G_r – ν_{peak} plane, with no clear correlation in this sample. A similar lack of correlation was reported for the 2FGL sample by [Linford et al. \(2012\)](#), promoting that HBLs are different from LBLs in the G_r – ν_{peak} plane. In this sample, 285 HBLs dilute the effects of the sample bias, and we proposed an explanation of the flattened trend in HBLs with the KN effect: as the characteristic electron and photon energies increase, the IC scattering becomes less efficient, suppressing the γ -ray emission and leading to the observed flattening of the relation.

3.3 Implications for Klein-Nishina Effects

In the standard one-zone leptonic framework for interpreting the broadband SEDs of HBLs, when IC scattering occurs in the Thomson regime, the SSC peak frequency follows ([Tavecchio et al. 1998](#); [Chen 2018](#); [Fan et al. 2023](#); [Xiao et al. 2025](#)),

$$\nu_{\text{SSC}} = \frac{4}{3} \gamma_p^2 \nu_{\text{syn}}, \quad (2)$$

where γ_p is the Lorentz factor of electrons that dominate the synchrotron emission. This relation implies a linear dependence between $\log \nu_{\text{SSC}}$ and $\log \nu_{\text{syn}}$, i.e., $\log \nu_{\text{SSC}} = \log \nu_{\text{syn}} + \text{const.}$, with the constant determined by γ_p . However, [Xiao et al. \(2025\)](#) found that the correlation between $\log \nu_{\text{SSC}}$ and $\log \nu_{\text{syn}}$ has a slope of ~ 0.64 for

HBLs, deviating from the Thomson expectation. This deviation is explained by the KN effect, which becomes important when the photon energy in the electron rest frame approaches the electron rest-mass energy. In this regime, the scattering cross-section is suppressed, reducing the efficiency of the IC process and weakening the γ -ray emission.

The onset of the KN regime can be approximately described by ([Tavecchio et al. 1998](#)),

$$\gamma_b \nu'_{\text{syn}} \gtrsim \frac{3}{4} \frac{m_e c^2}{h}, \quad (3)$$

where m_e is the electron mass, c is the speed of light, and h is the Planck constant, γ_b is the break Lorentz factor of the electron distribution, and ν'_{syn} is the synchrotron peak frequency in the co-moving frame, related to the observed synchrotron peak frequency ν_{syn} through $\nu_{\text{syn}} = \nu'_{\text{syn}} \delta / (1 + z)$. In the one-zone model, the synchrotron frequency averaged over the spectral shape for an electron of Lorentz factor γ_b is ([Tavecchio et al. 1998](#)),

$$\nu_{\text{syn}} = 3.7 \times 10^6 \gamma_b^2 B \frac{\delta}{1 + z}, \quad (4)$$

where B is the magnetic field strength (G) and z is the redshift. Eliminating the γ_b by combining Eqs. (3) and (4),

$$\nu_{\text{syn}} \gtrsim \nu_{\text{syn}}^c = 3.17 \times 10^{15} B^{1/3} \frac{\delta}{1 + z}, \quad (5)$$

which indicates that sources above this threshold are expected to experience significant KN suppression of the SSC emission.

[Xiao et al. \(2025\)](#) analysed the averaged broadband SEDs of *Fermi* BL Lacs and derived the critical ν_{syn}^c . They found that 359 out of 513 BL Lacs are affected by KN suppression. A large fraction of HBLs operate in a regime where IC scattering is significantly less efficient.

This provides a natural framework for interpreting the observed G_r – ν_{peak} relation. As discussed in the previous section, HBLs exhibit a much flatter distribution compared to LBLs and IBLs. Given their high ν_{peak} values (note that $\nu_{\text{syn}} = \nu_{\text{peak}}$ in this work), HBLs are more likely to satisfy the KN condition, leading to reduced IC efficiency and suppressed γ -ray emission. As shown in Fig. 4, the values of $\log G_r$ reach their plateau in a range $\log(\nu_{\text{peak}}/\text{Hz}) = 15.5 - 16$, and this behaviour may indicate the onset of KN suppression, marking the transition from the Thomson to the KN scattering regime in these sources. If this feature is indeed associated with KN effects, it can be used to constrain the magnetic field strength in the emitting region through Eq. (5).

In the model involving a shock in a relativistic jet, the γ -ray emission is produced in the innermost regions of the relativistic jet, whereas the radio emission originates from the outer region ([Marscher et al. 2008](#)). Within the SSC framework, the radio and γ -ray emissions are connected through the same population of relativistic electrons, with synchrotron photons serving as seed photons for IC scattering ([Finke et al. 2008](#)). The VLBI core emissions trace the compact, Doppler-boosted inner jet ([Deller & Middelberg 2014](#); [Böck et al. 2016](#); [Petrov & Kovalev 2025](#)). Therefore, the ratio between the γ -ray and VLBI radio emissions provides a valuable probe of the relativistic beaming effect and the origin of high-energy radiation. Moreover, the radio Doppler factors reported by [Liodakis et al. \(2018\)](#) cover a range comparable to the γ -ray Doppler factors derived by [Zhang et al. \(2002, 2020\)](#); [Chen et al. \(2024\)](#), further supporting that the compact radio emission is also closely related to the strong beaming effect. Since δ and B are inversely proportional and δ can vary widely (e.g., [Hovatta et al. 2009](#); [Liodakis et al. 2018](#); [Chen 2018](#); [Ye & Fan 2021](#)), δ could introduce substantial uncertainty in the estimation of the magnetic field. [Lian et al. \(2026\)](#) obtained the

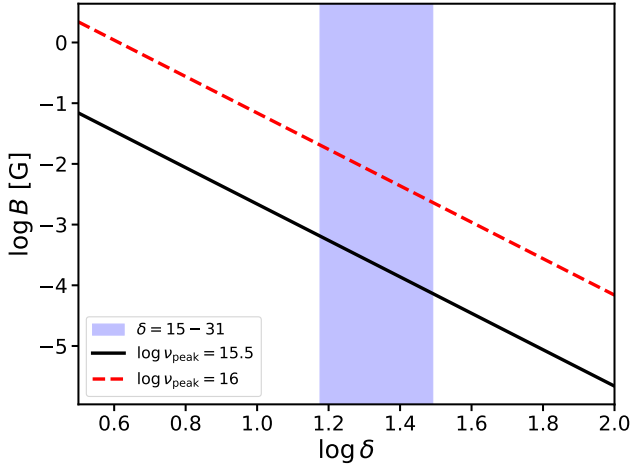


Figure 5. Magnetic field strength (B) versus Doppler factor (δ). The blue shaded region indicates the range $15 < \delta < 31$. Black solid and red dashed lines denote $\log \nu_{\text{peak}} = 15.5$ and 16 , respectively.

Doppler factors for 25 EHSPs in a range of $\delta = 5$ and $\delta = 30$ with an average of $\langle \delta \rangle = 14.6$. The average Doppler factor for *Fermi* BL Lacs is about $\langle \delta \rangle = 15$ in Zhang et al. (2020) and Chen et al. (2024). Tavecchio et al. (2010) obtained δ in a range of 20 to 30 for *Fermi* BL Lacs from the one-zone SSC model, but some extreme BL Lacs require $\delta > 40$. Tavecchio et al. (1998) investigated the magnetic field strength and Doppler factors of three TeV BL Lacs within a homogeneous SSC framework, taking the KN effect into account. For Mrk 421, they constrained the Doppler factor from the γ - γ pair-production opacity condition ($\delta > 15$; their Eq. [23]), while a more stringent KN condition gives an upper limit of $\delta_{\text{KN}} < 31$ (their Eq. [17]).

Motivated by these constraints, we explored the relation between the magnetic field strength B and the Doppler factor δ for HBLs with synchrotron peak frequencies in the range $\log \nu_{\text{peak}} = 15.5$ – 16 , assuming a redshift of $z = 0.3$ for BL Lacs (Tavecchio et al. 2010; Ajello et al. 2020, 2022). The resulting B - δ relation is shown in Fig. 5, where the blue shaded region corresponds to the range $15 < \delta < 31$. Although the exact onset of the KN suppression is difficult to determine observationally, a tentative plateau of G_r around $\log \nu_{\text{peak}} = 15.5$ – 16 can be seen in the HBL sample. Under this scenario, for sources ($z = 0.3$) with $\log \nu_{\text{peak}}$ between 15.5 and 16, the inferred magnetic field strength spans from $\log B = -4.14$ G to $\log B = -1.69$ G.

A small population of four HSP FSRQs was identified by Ajello et al. (2020). These sources exhibit hard γ -ray spectra similar to those of HBLs, but are located at higher redshifts. In general, *Fermi* FSRQs span a much broader redshift range than BL Lacs (Ajello et al. 2020, 2022). According to Eq. (5), the inferred magnetic field strength scales as $B \propto (1+z)^3$, making it highly sensitive to redshift. For the same Doppler factor, changing the redshift from $z = 0.3$ to $z = 1.8$ increases the inferred magnetic field by one order of magnitude. Therefore, high-redshift HSP FSRQs entering the KN regime are expected to possess stronger magnetic fields than BL Lacs. This is consistent with the SED fitting that higher magnetic fields are more common in FSRQs than BL Lacs (Ghisellini et al. 1998, 2010; Chen 2018; Fan et al. 2023).

Tavecchio et al. (2010) analysed the BL Lacs in the *Fermi* sample, including sources also detected at very high energies by

Cherenkov telescopes. They found that high-energy BL Lacs can be broadly divided into two groups. The majority are well described by a standard one-zone SSC model with moderate magnetic fields and electron energies, whereas EHBLs require unusually low magnetic fields and extremely energetic electrons, highlighting the diversity of physical conditions of the emission region in BL Lacs. Fan et al. (2023) derived magnetic fields for 1141 BL Lacs within a one-zone leptonic framework, finding an average value of $\log B = -0.51 \pm 1.66$. Similarly, Xiao et al. (2025) obtained an average value of $\log B = 0.23 \pm 0.33$ for *Fermi* BL Lacs based on physically constrained broadband SED modelling. Zhao et al. (2024) also studied the SEDs with the *JetSet* for a sample of 348 *Fermi* HBLs and computed that HBLs have the magnetic field in a range of $\log B = -3.66$ G to $\log B = 0.60$ G with an average of $\langle \log B \rangle = -1.25$ G. Láinez et al. (2025) identified 66 new EHSP candidates and calculated the magnetic field strength based on a one-zone synchrotron/SSC framework. They proposed that ν_{peak} reflects a more complex combination of parameters (including γ_b , B , and δ) and may capture broader physical conditions in the jet. However, they showed that sources with the lower ν_{peak} have the lower magnetic field (Fig. 6 of Láinez et al. 2025). As discussed in Láinez et al. (2025), when the synchrotron peak shifts to higher frequencies, the energy stored in the magnetic field becomes comparable to that of the relativistic electrons, supporting a more balanced and energetically efficient jet environment in most EHBLs. Lian et al. (2026) computed the magnetic field strength ranging from $\log B = -2$ G to $\log B = -0.22$ G with an average of $\langle \log B \rangle = -1.15$ G for 25 EHSPs.

Our estimated magnetic field strengths are broadly consistent with previous studies. The KN effect plays an important role in shaping the γ -ray loudness, particularly for HBLs. This effect can explain the flattening of the G_r - ν_{peak} relation and the absence of a significant correlation in the 2FGL sample reported by Linford et al. (2012), where a larger number of HBLs were included in the analysis of the G_r and ν_{peak} .

4 CONCLUSIONS

In this work, we investigated the relation between radio and γ -ray properties for a large sample of 1687 *Fermi* blazars, including an investigation of the relation between the ν_{peak} and the G_r , as well as its implications for high-energy processes. The main conclusions are as follows:

(i) Based on a large sample of 1687 *Fermi* blazars, we presented a clear positive correlation between radio and γ -ray fluxes for both BL Lacs and FSRQs. This result suggests that SSC emission contributes to the high-energy emission in both subclasses, providing a natural link between the radio and γ -ray bands.

(ii) Compton cooling would be expected to produce a positive correlation between G_r and Γ_γ . However, an anti-correlation is observed for the sample of 1687 *Fermi* blazars. This behaviour is instead interpreted as a consequence of the SED shift, rather than being driven by Compton cooling. Therefore, the shift of the SED appears to play a dominant role in shaping the G_r - Γ_γ relation.

(iii) The relation between G_r and ν_{peak} exhibits distinct behaviours among different blazar subclasses. FSRQs and LBLs show a weak positive trend, contributed from SSC emissions, while the substantial scatter is likely associated with EC emissions. In contrast, HBLs display a much flatter distribution. We interpreted this flattening as a consequence of the reduced IC efficiency in the KN regime. Assuming $\delta \in (15, 31)$ in the SSC leptonic framework, the onset

of KN suppression around $\log(\nu_{\text{peak}}/\text{Hz}) \sim 15.5\text{--}16$ allows one to constrain the magnetic field strength of $-4.14 < \log(B/\text{G}) < -1.69$ for those HBLs transitioning from the Thomson to the KN regime.

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DATA AVAILABILITY

The data underlying this article are publicly available. The radio data were obtained from the RFC VLBI catalogue (DOI: 10.25966/dhrk-zh08), and the γ -ray data were obtained from the *Fermi* 4LAC-DR3 AGN catalogue (DOI: <https://doi.org/10.26093/cds/vizier.22630024>). Additional data products generated during this study are available from the corresponding author upon reasonable request. The complete Table 1 underlying this article is available in Zenodo, at <https://doi.org/10.5281/zenodo.21695209> [doi].

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